

Week 1 Deliverable: Member Conversion Prediction Model

IAASSE Data Science Project — Target Definition & Data Collection

Date: August 30, 2026 | **Team:** Team 2–4 | **Tools:** Python, Pandas, NumPy, Matplotlib, Seaborn

Executive Summary

This report presents the foundation for a machine learning model that predicts which free-tier IAASSE members will convert to paid membership. The prediction target is a binary classification: converted = 1 if a member upgrades within 90 days of signup, 0 otherwise. All predictor features are strictly derived from the member's first 30 days of activity to prevent data leakage and ensure real-time deployability. A synthetic dataset of **4,433 eligible members** was generated, validated, and explored. The dataset exhibits a **21.9% conversion rate** with a **3.6:1 class imbalance**. Early exploratory analysis reveals strong predictive signals: Managers convert at **37.2%** (3.8× Students at 9.9%), members in developed countries (Germany 24.3%) outperform emerging markets (Brazil 15.7%), and converted members show consistently higher engagement across all metrics in their first 30 days. No data quality issues or leakage risks were detected.

Problem Definition

Business Context

IAASSE operates a freemium membership model. Free-tier members receive limited access to resources, events, and community features. Converting these members to paid plans is critical to revenue growth. Currently, outreach is untargeted — all free members receive the same marketing. A predictive model would enable **prioritized, personalized outreach** to members most likely to convert, improving marketing ROI and conversion rates.

Prediction Target

Attribute	Specification
Problem Type	Binary Classification
Target Variable	converted
Positive Class (1)	Member upgraded to paid status within 90 days of free-tier registration
Negative Class (0)	Member did not upgrade within 90 days
Observation Window	First 30 days of member activity
Prediction Horizon	90 days from signup
Eligibility Criteria	Members who signed up ≥ 90 days before analysis date (2026-08-30)

Anti-Leakage Protocol

Data leakage — using future information to predict the past — is the most common failure mode in conversion modeling. The following safeguards were implemented:

Safeguard	Implementation
Temporal cutoff	All engagement metrics capped at first 30 days
Eligibility filter	Members with < 90 days since signup removed (outcome unknown)
Feature audit	Every feature reviewed to confirm it would be available at prediction time
No post-conversion features	days_since_signup is the only temporal feature; no upgrade date or plan type included

Methodology

Data Source

Synthetic dataset generated to simulate realistic IAASSE membership data with business-logic-driven conversion patterns.

Inclusion Criteria

- Registered for a free-tier IAASSE account
- Signup date is at least 90 days before analysis date (ensures known outcome)
- Has at least partial engagement data from first 30 days

Exclusion Criteria

- Members who signed up within the last 90 days (outcome still unknown)
- Members placed directly on paid plans at registration (not free-tier conversions)
- Duplicate or test accounts

Data Generation Approach

The generation process used a two-stage approach to prevent leakage:

1. **Stage 1 — Demographic-driven probability:** Conversion probability was computed from demographic features (job title, age, country) using realistic effect sizes derived from SaaS membership benchmarks.
2. **Stage 2 — Engagement correlation:** For members assigned as “converted,” engagement metrics (logins, downloads, webinars, posts) were statistically boosted to reflect the known correlation between early engagement and conversion — without accessing post-conversion data.

This approach ensures the dataset exhibits realistic patterns while maintaining strict temporal integrity.

Dataset Overview

Metric	Value
Total Records Generated	5,000 members

Metric	Value
After Eligibility Filter (≥ 90 days)	4,433 members
Date Range	90–729 days since signup
Total Features	18
Modeling Features	16
Target Variable	1 (converted)
ID Column	1 (member_id)
Converters (1)	970 (21.88%)
Non-Converters (0)	3,463 (78.12%)
Class Imbalance Ratio	3.6 : 1

Data Dictionary

#	Feature	Type	Description	Modeling Role
1	member_id	String	Unique member identifier	ID — exclude
2	age	Integer (18–64)	Member age in years	Feature
3	country	Categorical (8 levels)	Country of residence	Feature
4	job_title	Categorical (8 levels)	Job title / role	Feature
5	days_since_signup	Integer (90–729)	Days since registration	Feature — evaluate

#	Feature	Type	Description	Modeling Role
6	login_frequency_30d	Integer (1–20)	Platform logins in first 30 days	Feature
7	resources_downloaded	Integer (0–14)	Resources downloaded in first 30 days	Feature
8	webinars_attended	Integer (0–8)	Webinars attended in first 30 days	Feature
9	forum_posts	Integer (0–12)	Forum posts in first 30 days	Feature
10	profile_completion_pct	Float (0.7–100.0)	Profile completion percentage	Feature
11	events_registered	Integer (0–10)	Events registered for	Feature
12	events_attended	Integer (0–8)	Events actually attended	Feature
13	newsletter_opens	Integer (0–14)	Newsletter emails opened	Feature
14	avg_session_duration_min	Float (0.4–104.3)	Average session duration in minutes	Feature
15	support_tickets_raised	Integer (0–6)	Support tickets raised	Feature
16	emails_to_support	Integer (0–4)	Emails sent to support	Feature
17	referral_cou	Integer	Members referred	Feature

#	Feature	Type	Description	Modeling Role
	nt	(0–4)		
18	converted	Binary (0/1)	Target: upgraded within 90 days	Target

Feature Categories

- **Demographic/Static:** 3 features (age, country, job_title)
- **Behavioral/Engagement:** 10 features (logins, downloads, webinars, posts, profile completion, events registered/attended, newsletter opens, session duration, referrals)
- **Support/Communication:** 3 features (support tickets, emails to support, referral count)
- **Temporal:** 1 feature (days_since_signup)
- **Target:** 1 feature (converted)

Data Quality Report

Validation Checks

#	Check	Status	Detail
1	Duplicate Member IDs	PASS	0 duplicates found across 4,433 records
2	Missing Values	PASS	0 missing values in all 18 columns
3	Future Dates	PASS	All days_since_signup ≤ 729 (analysis date is 2026-08-30)
4	Range Validation — Age	PASS	18–64 years
5	Range Validation — Engagement	PASS	All counts ≥ 0; maximums within

#	Check	Status	Detail
			realistic bounds
6	Range Validation — Percentages	PASS	profile_completion_pct: 0.7%–100.0%
7	Eligibility Criteria	PASS	All members have ≥ 90 days since signup (min: 90, max: 729)
8	Data Leakage	PASS	No post-conversion features; all modeling features from first 30 days only
9	Class Balance	WARNING	3.6:1 imbalance — mitigation planned for Week 3

Numeric Features Summary

Feature	Mean	Std Dev	Min	25%	50%	75%	Max
age	41.2	13.5	18	30	41	53	64
days_since_signup	414.6	185.2	90	251	421	577	729
login_frequency_30d	8.0	2.9	1	6	8	10	20
resources_downloaded	5.0	2.2	0	3	5	6	14
webinars_attended	2.0	1.4	0	1	2	3	8
forum_posts	3.0	1.7	0	2	3	4	12
profile_com	66.7	23.4	0.7	50.4	70.5	86.4	100.0

Feature	Mean	Std Dev	Min	25%	50%	75%	Max
pletion_pct							
events_registered	3.0	1.8	0	2	3	4	10
events_attended	1.8	1.3	0	1	2	3	8
newsletter_opens	6.0	2.1	0	5	6	7	14
avg_session_duration_min	19.9	14.1	0.4	9.6	16.7	26.6	104.3
support_tickets_raised	1.0	1.0	0	0	1	2	6
emails_to_support	0.5	0.7	0	0	0	1	4
referral_count	0.3	0.5	0	0	0	1	4

Key Findings

Conversion by Job Title

Job Title	Conversion Rate	Sample Size	vs. Overall
Manager	37.2%	510	+15.3 pp
Consultant	30.3%	393	+8.4 pp
Professor	26.1%	230	+4.2 pp
Data Analyst	22.7%	819	+0.8 pp
Researcher	22.4%	540	+0.5 pp
Freelancer	22.3%	228	+0.4 pp

Job Title	Conversion Rate	Sample Size	vs. Overall
Software Engineer	20.9%	997	-1.0 pp
Student	9.9%	716	-12.0 pp

Insight: Job title is a strong predictive signal. Managers convert at **3.8×** the rate of Students. Senior roles (Manager, Consultant, Professor) consistently outperform entry-level positions, aligning with budget authority and decision-making power.

Conversion by Country

Country	Conversion Rate	vs. Overall
Germany	24.3%	+2.4 pp
Canada	24.2%	+2.3 pp
Australia	23.1%	+1.2 pp
USA	22.9%	+1.0 pp
UK	21.7%	-0.2 pp
India	20.3%	-1.6 pp
Nigeria	18.6%	-3.3 pp
Brazil	15.7%	-6.2 pp

Insight: Members in developed countries (Germany, Canada, Australia, USA) show higher conversion rates than emerging markets (Brazil, Nigeria). This may reflect differences in disposable income, organizational training budgets, or IAASSE's market penetration.

Engagement Patterns (First 30 Days)

Metric	Converted Mean	Not Converted Mean	Difference	Lift
Logins	8.5	7.9	+0.6	+8%
Resources	5.5	4.9	+0.6	+12%

Metric	Converted Mean	Not Converted Mean	Difference	Lift
Downloaded				
Webinars Attended	2.3	1.9	+0.4	+21%
Forum Posts	3.2	2.9	+0.3	+8%

Insight: Webinar attendance shows the strongest discriminatory power (+21% lift). Members who attend educational content demonstrate higher intent to convert. Resource downloads also show meaningful lift (+12%), suggesting content consumption is a strong buying signal.

Profile Completion & Session Behavior

Metric	Converted Mean	Not Converted Mean	Lift
Profile Completion (%)	72.4	65.0	+11%
Avg Session Duration (min)	22.3	19.2	+16%
Events Attended	2.1	1.7	+24%
Newsletter Opens	6.4	5.9	+8%

Insight: Event attendance (+24% lift) and session duration (+16% lift) are strong behavioral signals. Members who invest time in the platform (longer sessions, attending events, completing profiles) are significantly more likely to convert.

Visualizations

All findings are supported by the 6-panel EDA chart (week1_eda.png), which includes:

1. **Target Distribution** — Pie chart showing 21.9% conversion rate
2. **Conversion by Job Title** — Horizontal bar chart across 8 roles
3. **Conversion by Country** — Horizontal bar chart across 8 countries
4. **Engagement Comparison** — Grouped bar chart comparing converted vs. non-converted across 4 metrics
5. **Profile Completion Distribution** — Overlapping histograms showing density differences

6. **Age Distribution** — Overlapping histograms by conversion status

Limitations & Risks

#	Risk	Impact	Likelihood	Mitigation Strategy
1	Class imbalance (3.6:1)	Model biases toward predicting “not converted,” poor recall for converters	High	Use <code>class_weight='balanced'</code> , SMOTE oversampling, or threshold tuning in Week 3
2	Synthetic data	May not perfectly reflect real-world IAASSE member behavior	Medium	Validate findings with real data when available; adjust feature weights accordingly
3	Feature leakage (unintentional)	Inflated test performance, model fails in production	Low	Strict 30-day observation window; audit all features before modeling
4	Categorical cardinal	High-dimensional sparse	Low	One-hot encoding for low-cardinality features (8 job titles, 8 countries)

#	Risk	Impact	Likelihood	Mitigation Strategy
	ality	features after encoding		
5	Tempo ral drift	Model performan ce degrades as member behavior evolves	Medium	Monitor model performance monthly; plan retraining every quarter

Reproducibility

Environment

Component	Version
Python	3.10+
Pandas	2.3.2
NumPy	2.2.5
Matplotlib	3.9.0
Seaborn	0.13.2

Random Seeds

All random operations use fixed seed `np.random.seed(42)` to ensure identical results across runs.

Files Required to Reproduce

File	Purpose
<code>iaasse_membership_data.csv</code>	Input dataset (4,433 × 18)

File	Purpose
data_dictionary.csv	Feature documentation
week1_eda.png	Generated visualizations
week1_complete_notebook.py	Full code with 8 sequential cells